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Optimizing ER Biomarker Expression Classification with Hybrid Transformers: Multi-Color Space Ensemble and Fusion Approach

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Abstract. Breast cancer remains one of the most prevalent and life-threatening diseases globally, where accurate assessment of Immunohistochemistry (IHC) Estrogen Receptor (ER) biomarker expression is crucial for guiding treatment decisions. However, manual evaluation of IHC-stained histopathological images is prone to inter-observer variability, leading to inconsistencies in diagnosis. While Deep learning has shown promise in automating ER classification, optimizing model performance and generalization across diverse imaging conditions remains a challenge. This study explores the effectiveness of two hybrid transformer models—Swin Transformer V2 and EdgeNeXt across three color spaces (RGB, HSL, and LAB) to determine the optimal model-color space combination for IHC ER biomarker classification. Using Multi-Criteria Decision Analysis (MCDA) with the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) method, models are ranked based on accuracy, precision, recall, and F1-score on both test and out-of-distribution (OOD) sets. The top-performing models are further optimized through ensemble learning (Confidence Weighted Average Ensemble) and fusion strategies (Cross Attention Fusion). Results show that the Confidence-Weighted Average Ensemble achieves the highest classification accuracy (0.9054 on test and 0.9228 on OOD set), outperforming individual models and fusion methods, demonstrating superior generalization and robustness to staining variations. Additionally, Grad-CAM visualizations enhance interpretability by identifying key histological regions influencing model predictions. This study analyzes transformer-based models across multiple color spaces and demonstrates the efficacy of ensemble learning in enhancing IHC ER biomarker classification.

Keywords: ER expression, Immunohistochemistry (IHC), Hybrid Transformer, Confidence Weighted Average Ensemble, Cross Attention Fusion, Breast Cancer

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1 Introduction

Immunohistochemistry (IHC) plays a critical role in breast cancer diagnosis and treatment planning by evaluating Estrogen Receptor (ER) biomarker expression, which directly influences therapeutic decisions [1]. Accurate classification of ER expression is essential for personalized treatment; however, manual evaluation by pathologists is subjective, time-intensive, and prone to variability, making consistency a challenge [2]. Machine learning (ML) and Deep learning (DL) models have shown significant potential in automating ER classification by improving accuracy and reducing observer dependency [3–6]. Despite these advancements, achieving generalizability across diverse staining conditions, tissue variations, and imaging protocols remains a key challenge.

To address these limitations, this study investigates hybrid transformer models Swin Transformer [7] and EdgeNeXt [8] trained on three distinct color spaces (RGB, HSL, and LAB) for ER expression classification. The objective is to identify the optimal model-color space combination for robust and accurate classification. Given the variation in model performance across datasets, we employ Multi-Criteria Decision Analysis (MCDA) using the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) [9] to systematically rank models based on their classification performance on both test and out-of-distribution (OOD) sets. To further improve classification accuracy and generalization, ensemble learning (Confidence Weighted Average Ensemble) and fusion strategies (Cross Attention Fusion) are introduced, leveraging the complementary strengths of individual models. Performance is rigorously evaluated using Accuracy, Precision, Recall, and F1-score, ensuring a comprehensive assessment of classification reliability. Beyond performance metrics, interpretability is crucial for clinical applications. This study introduces Modified Grad-CAM approaches, including Confidence-Based Weighted Dynamic Grad-CAM, to enhance biomarker localization and improve understanding of ensemble decision-making. Confidence-Based Weighted Grad-CAM assigns greater influence to models with higher confidence, resulting in more precise visualization of critical histological regions influencing classification outcomes.

Through a comparative analysis of individual models, ensemble learning, and fusion techniques, this study establishes a robust framework for automated IHC ER biomarker expression level classification. The findings contribute to the development of generalizable and interpretable deep-learning models, ultimately assisting pathologists in achieving more accurate and efficient breast cancer diagnosis.

2 Related Work

Several studies have explored the color space transformations on improving classification performance using both ML and DL techniques. Mouelhi et al. [3] used HSL saturation (S) and Fisher Linear Discriminant (FLD) to distinguish positive and negative/benign nuclei in ER. Mungle et al. [4] applied an MRF-ANN

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approach in RGB to classify and score ER expression in breast cancer IHC images. Tewary et al. [5] introduced an IHC profiler for ER and PR assessment, leveraging CMYK for proportion analysis and ML models for intensity classification. Mouelhi et al. [13] proposed an ER classification method using enhanced RGB images, applying adaptive thresholding and segmentation. Ahmad et al. [12] automated ER classification in IHC-stained WSIs using DenseNet, achieving 56.8% concordance with manual evaluation and 81.1% alignment with treatment decisions. Jamaluddin et al. [14] used the HSV color space with adaptive thresholding and post-processing for cell detection and PN classification, achieving 95% precision and 91% recall in ER-positive and ER-negative cell classification. Saha et al. [16] introduced HscoreNet, a deep learning framework inspired by SegNet and U-Net, achieving 95.87% precision and 94.53% accuracy in ER and PR biomarker analysis. Hasan et al. [10] developed a real-time system for whole-slide image segmentation and classification, integrating SegNet, U-Net, ResNet, VGG, and DenseNet, achieving an F1-score of 0.902 for segmentation and 0.833 for classification. Feng et al. [11] used DenseNet for nuclear detection and ER/PR classification in ER-IHC-stained images, achieving an F1-score of 0.855 for detection and 0.8306 for classification. Mahanta et al. [6] developed IHC-Net for automated nuclear segmentation, extracting color and texture features, and classifying using an ensemble of Naïve Bayes, SVM, and Random Forest. Fauzi et al. [12] introduced a deep learning-based Allred scoring system integrating adaptive thresholding and color deconvolution, achieving 82.5% agreement with pathologists. Jamaluddin et al. [15] developed a CNN-based classifier for ER-stained WSIs, achieving 88.8% accuracy and 97.5% AUC. Torres et al. [17] applied deep learning-based instance segmentation using SOLOv2 with a ResNet-50 backbone for precise cell separation in breast cancer IHC images.

These findings underscore the crucial role of color space selection and model choice in improving ER biomarker classification

3 Methodology

This study presents a comprehensive approach to ER biomarker expression classification by leveraging hybrid transformer models across three distinct color spaces to enhance classification accuracy and robustness. The study involves training Swin Transformer V2 and EdgeNeXt on three color spaces and evaluating their performance using MCDA with the TOPSIS [9] method to systematically rank and identify the most effective model and color space combination. Based on the TOPSIS ranking, the top three combinations were further optimized using Confidence Weighted Average Ensemble and Cross Attention Fusion to improve interpretability and classification accuracy. Performance was rigorously evaluated on both a standard test and an OOD set using Accuracy, Precision, Recall, and F1 Score. Additionally, Grad-CAM was employed to analyze the interpretability of both ensemble and fusion models, providing visual insights into their decision-making processes across both the test and OOD set. Figure 1 provides an overview of the proposed methodology.

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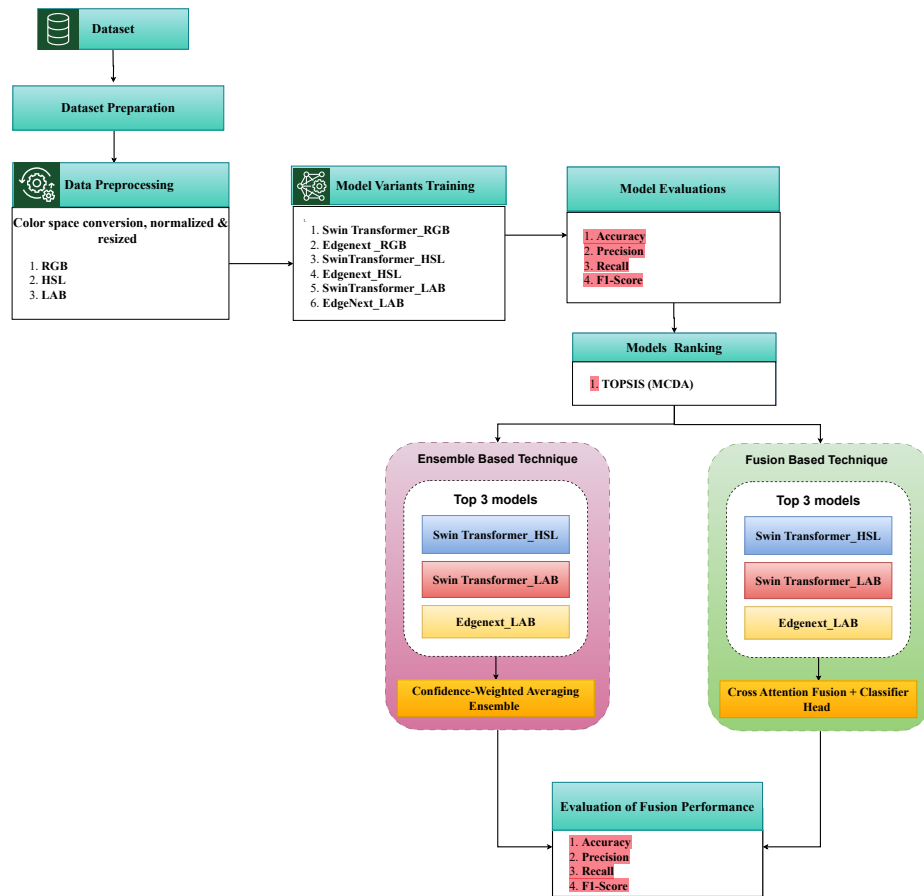


Fig. 1. Overview of the Proposed Methodology

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3.1 Dataset Details

This study utilized two datasets: a publicly available dataset and an in-house dataset, to ensure comprehensive evaluation and generalization of the proposed methodology.

- **ER AIDPATH Dataset:** Part of the AIDPATH project [18], this dataset includes 50 breast cancer WSIs scanned with an Aperio CS at $40\times$ magnification. ER IHC-stained samples were patchified into 1024×1024 pixel tiles and labeled into four ER expression levels—Negative, Low, Moderate, or High—verified by a pathologist.
- **ER_Micro Dataset:** The in-house dataset was developed with support from the North East Cancer Hospital and Research Institute (NECHRI) in Guwahati, Assam, India. It comprises 40 ER IHC-stained slides captured with a Leica microscope at $40\times$ magnification (1920×1080 resolution). Used as an OOD dataset, to assess model generalization under varying staining and imaging conditions.

Dataset Preprocessing The dataset was split into Training, Validation, Test, and OOD sets as shown in Table 1, with class imbalance noted in Low and High Expression categories. To address this, horizontal and vertical flipping were applied, resulting in 3,000 samples per class (12,000 total in Training). Preprocessing included conversion to RGB, HSL, and LAB color spaces to assess their impact on model performance is shown in Figure 2. All images were resized to 224×224 pixels for consistency.

Table 1. Dataset Distribution Across Training, Validation, Test, and OOD Sets

Class	Training	Validation	Test	OOD
Negative Expression	3000	346	508	226
Low Expression	1843	204	227	113
Moderate Expression	3000	333	377	208
High Expression	2071	230	255	525

3.2 Model Selection

To systematically evaluate the influence of color spaces on IHC ER biomarker expression classification, two hybrid transformer models are employed —Swin TransformerV2 [7] and EdgeNeXt [8] were trained separately on three distinct color spaces: RGB, HSL and LAB. This approach enabled a comprehensive assessment of how different color representations impact feature extraction and classification performance. Each color space offers unique advantages for ER

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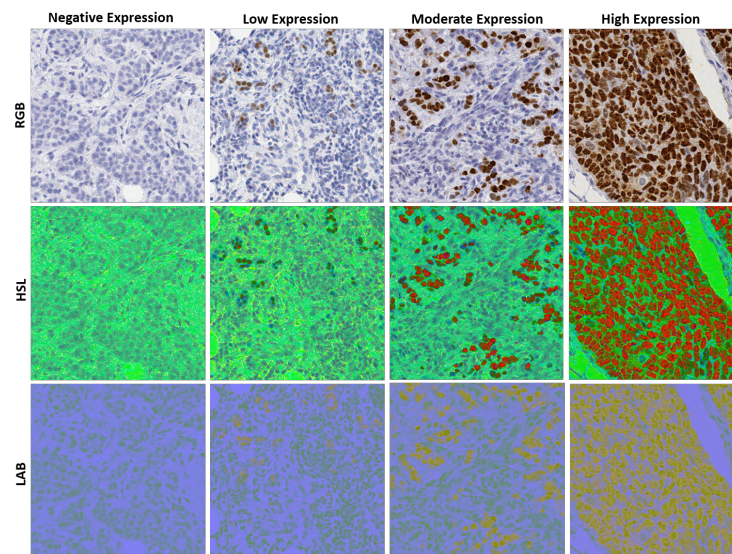


Fig. 2. Color transformation into RGB, HSL and LAB

image analysis. RGB preserves raw color information and serves as the standard in deep learning applications. Lab separates luminance (L) from chromatic components (A^* and B^*), enhancing robustness to illumination variations and improving color-based feature learning. HSL represents colors in terms of hue, saturation, and lightness, capturing texture and contrast variations crucial for biomarker expression analysis.

For each transformer model, individual variants were trained on these color spaces, resulting in 6 distinct models (2 models $\times$ 3 color spaces). These variants—Swin Transformer_RGB, Swin Transformer_Lab, Swin Transformer_HSL, EdgeNeXt_RGB, EdgeNeXt_Lab and EdgeNeXt_HSL, underwent rigorous evaluation to determine the optimal model-color space combinations.

3.3 Ensemble & Fusion Strategies

To enhance the robustness and generalization of the IHC ER biomarker expression level classification models, we employed two advanced strategies—Ensemble Learning and Fusion Techniques—were implemented using the top-performing model-color space combinations identified through MCDA-TOPSIS [9]. These approaches aim to leverage complementary strengths across models to improve classification accuracy and interpretability.

Confidence-Weighted Average Ensemble: Ensemble learning enhances classification performance by combining predictions from multiple models, reducing

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misclassification risk. This study employs a Confidence-Weighted Average Ensemble to improve the robustness of IHC ER biomarker classification by leveraging the strengths of the top-performing models identified through MCDA-TOPSIS. Unlike conventional ensemble methods, this approach assigns higher weights to models with superior performance, ensuring a more reliable classification. The top three models—Swin Transformer_HSL, Swin Transformer_Lab, and EdgeNeXt_Lab—were selected, with weight assignments based on **key evaluation metrics, including accuracy, precision, recall, and F1-score.** The final classification was obtained by aggregating the weighted probability outputs from each model, allowing stronger models to exert greater influence on the decision-making process, thereby improving classification stability. Mathematically, the Confidence Weighted Average Ensemble is defined as follows:

1. **Model Weight Assignment:** Each model M_i is assigned a confidence weight w_i , normalized as:

$$w_i = \frac{S_i}{\sum_{j=1}^3 S_j} \quad (1)$$

where S_i denotes the performance score of model M_i .

2. **Weighted Probability Aggregation:** The final classification probability is computed as:

$$P(y | x) = \sum_{i=1}^3 w_i \cdot P_i(y | x) \quad (2)$$

where $P_i(y | x)$ is the predicted probability of class y given input x by model M_i .

3. **Final Decision:** The predicted class $\hat{y}$ is determined by:

$$\hat{y} = \arg \max_y P(y | x) \quad (3)$$

This method effectively reduces individual model biases, minimizes weaker predictions, and enhances classification reliability by integrating complementary strengths of multiple color-space-based models.

Cross Attention Fusion: Cross Attention Fusion was employed to enhance IHC ER biomarker classification by integrating multiple color-space-based feature representations. Since different color spaces highlight distinct histopathological features—such as texture, contrast, and structural details—this fusion strategy enables a more comprehensive representation. The Cross Attention mechanism dynamically assigns importance to the most relevant features, ensuring an adaptive and discriminative representation.

Feature maps were extracted from Swin Transformer (HSL & LAB) and EdgeNeXt (LAB) models, each trained on different color-space representations. These feature maps were projected into a shared latent space using convolutional transformations for alignment. The Cross Attention mechanism then computed importance weights, enhancing informative regions while suppressing redundant

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features, leading to a structured and hierarchical integration. Mathematically, the feature fusion is defined as follows:

1. **Feature Projection:** Feature representations X^H, X^L, X^E from Swin Transformer_HSL, Swin Transformer_LAB, and EdgeNeXt_LAB models are projected into a common space:

$$Z^H = W_H * X^H, \quad Z^L = W_L * X^L, \quad Z^E = W_E * X^E \quad (4)$$

2. **Attention Weight Calculation:** Attention scores are computed using a softmax function:

$$A = \text{Softmax}(W_A * (Z^H + Z^L + Z^E)) \quad (5)$$

3. **Feature Aggregation:** The final fused representation is obtained as:

$$F = A_H \cdot X^H + A_L \cdot X^L + A_E \cdot X^E \quad (6)$$

4. **Classification Layer:** The fused representation F undergoes Global Average Pooling (GAP) and is classified using a softmax function:

$$Y = \text{Softmax}(W_F \cdot \text{GAP}(F)) \quad (7)$$

This fusion strategy adaptively weights feature contributions, ensuring a robust and discriminative classification of biomarker expression.

3.4 Experimental Setting and Model Training

All models in this study were trained from scratch without pre-trained weights to ensure domain-specific feature learning for IHC ER biomarker expression classification. Training was conducted using TensorFlow-Keras on an NVIDIA RTX 3060 GPU (12GB VRAM). Swin Transformer and EdgeNeXt models were trained separately on three color spaces (RGB, HSL, and LAB) using a standardized setup: the Adam optimizer with learning rate of 0.0001, a batch size of 32, and categorical cross-entropy as the loss function. Each model was trained for up to 40 epochs, with early stopping (patience = 5) based on validation loss to prevent overfitting.

4 Result and Discussion

This section analyzes the performance of three hybrid transformers across three color spaces using key metrics and MCDA-TOPSIS ranking to identify the best model-color space combination. Top models are further evaluated with ensemble and fusion techniques to enhance classification accuracy. Performance is tested on both standard and OOD sets for robustness. Misclassification analysis using confusion matrices provides insights into classification errors, and Grad-CAM visualizations reveal critical features influencing model predictions.

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4.1 Performance Evaluations and Misclassifications Analysis of Hybrids Transformer across different color spaces on Test and OOD Set

Tables 2 and 3 present the classification performance of different models across color spaces on the test and OOD sets, respectively. On the test set, Swin Transformer_RGB achieved the highest accuracy (0.9072), followed by Swin Transformer_HSL (0.8956) and Swin Transformer_LAB (0.8939), indicating that RGB and HSL color spaces effectively enhance feature representation. On the OOD set, Swin Transformer_HSL demonstrated the highest generalization capability (0.9178), followed by Swin Transformer_LAB (0.9070) and EdgeNeXt_LAB (0.9014).

Table 2. Classification Performance Across Different Color Spaces and Models on Test Set

Color Space	Model	Accuracy	Precision	Recall	F1 Score
RGB	Edgenext_RGB	0.8925	0.8863	0.8925	0.8892
	SwinTransformer_RGB	0.9072	0.8946	0.9072	0.8992
HSL	Edgenext_HSL	0.8813	0.8726	0.8813	0.8757
	SwinTransformer_HSL	0.8956	0.8956	0.8956	0.8950
LAB	EdgeNext_LAB	0.8937	0.8759	0.8937	0.8812
	SwinTransformer_LAB	0.8939	0.8770	0.8939	0.8822

Table 3. Classification Performance Across Different Color Spaces and Models on OOD Set

Color Space	Model	Accuracy	Precision	Recall	F1 Score
RGB	Edgenext_RGB	0.7701	0.8691	0.7645	0.7701
	SwinTransformer_RGB	0.8327	0.8433	0.8316	0.8327
HSL	Edgenext_HSL	0.8035	0.8330	0.7731	0.8035
	SwinTransformer_HSL	0.9178	0.9387	0.9253	0.9178
LAB	EdgeNext_LAB	0.9014	0.9097	0.9053	0.9014
	SwinTransformer_LAB	0.9070	0.9256	0.9142	0.9070

Figures 3 and 4 illustrate the confusion matrices for the test and OOD datasets. In the test set, the models achieve high classification accuracy for Negative, Low, and High ER expression levels in HSL and LAB color spaces, with minimal misclassification. However, Moderate expression is frequently misclassified as High or Low, highlighting the challenge of distinguishing subtle differences between these expression levels. In the OOD set, Swin Transformer performs well in HSL and LAB, achieving the best results across all expression levels but performs poorly in RGB, exhibiting a high misclassification rate. EdgeNeXt, on the other hand, shows lower performance in HSL, particularly in classifying Negative

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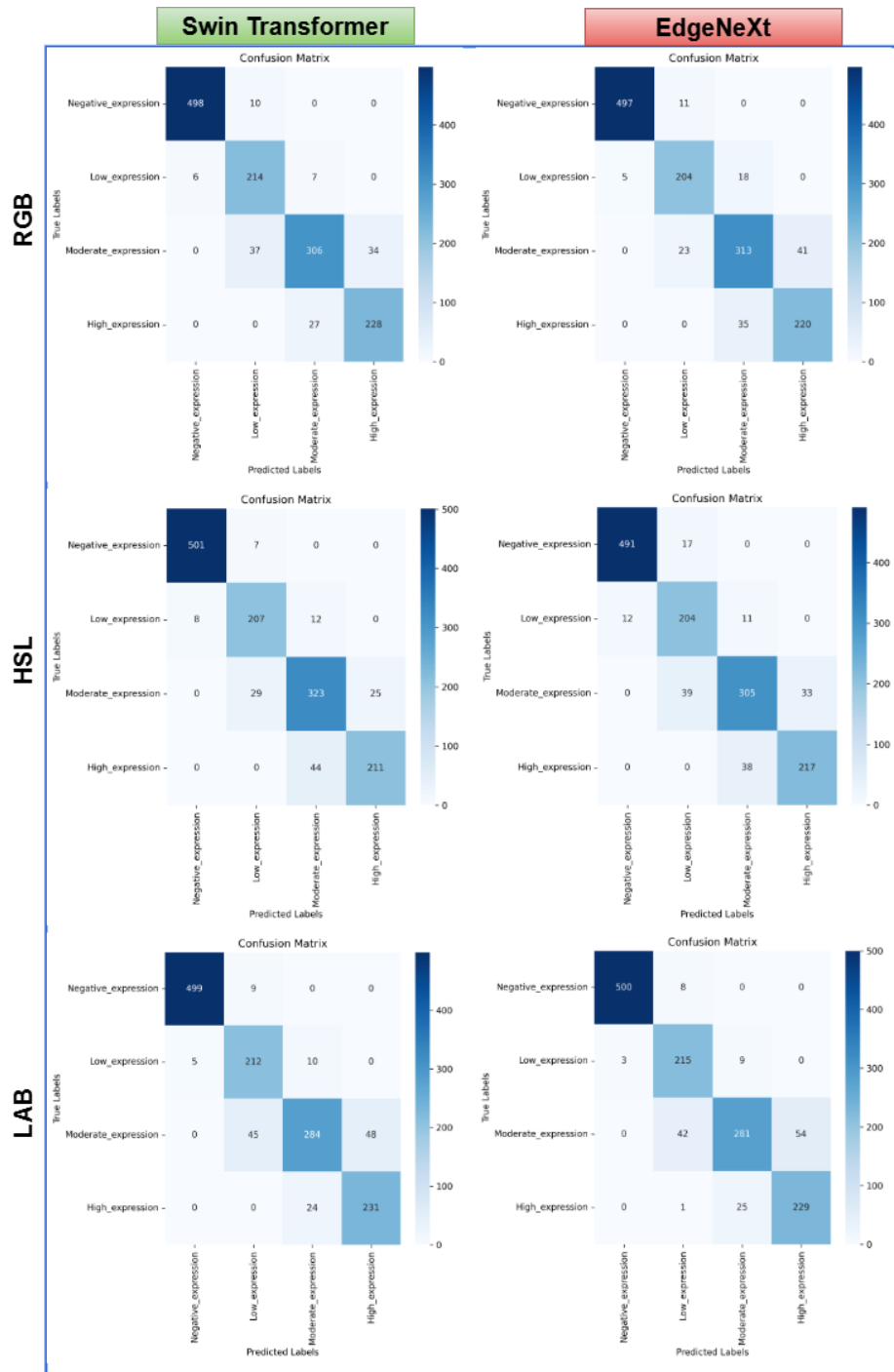


Fig. 3. Confusion matrices for Swin Transformer and EdgeNeXt in RGB, HSL, and LAB color spaces on the test set

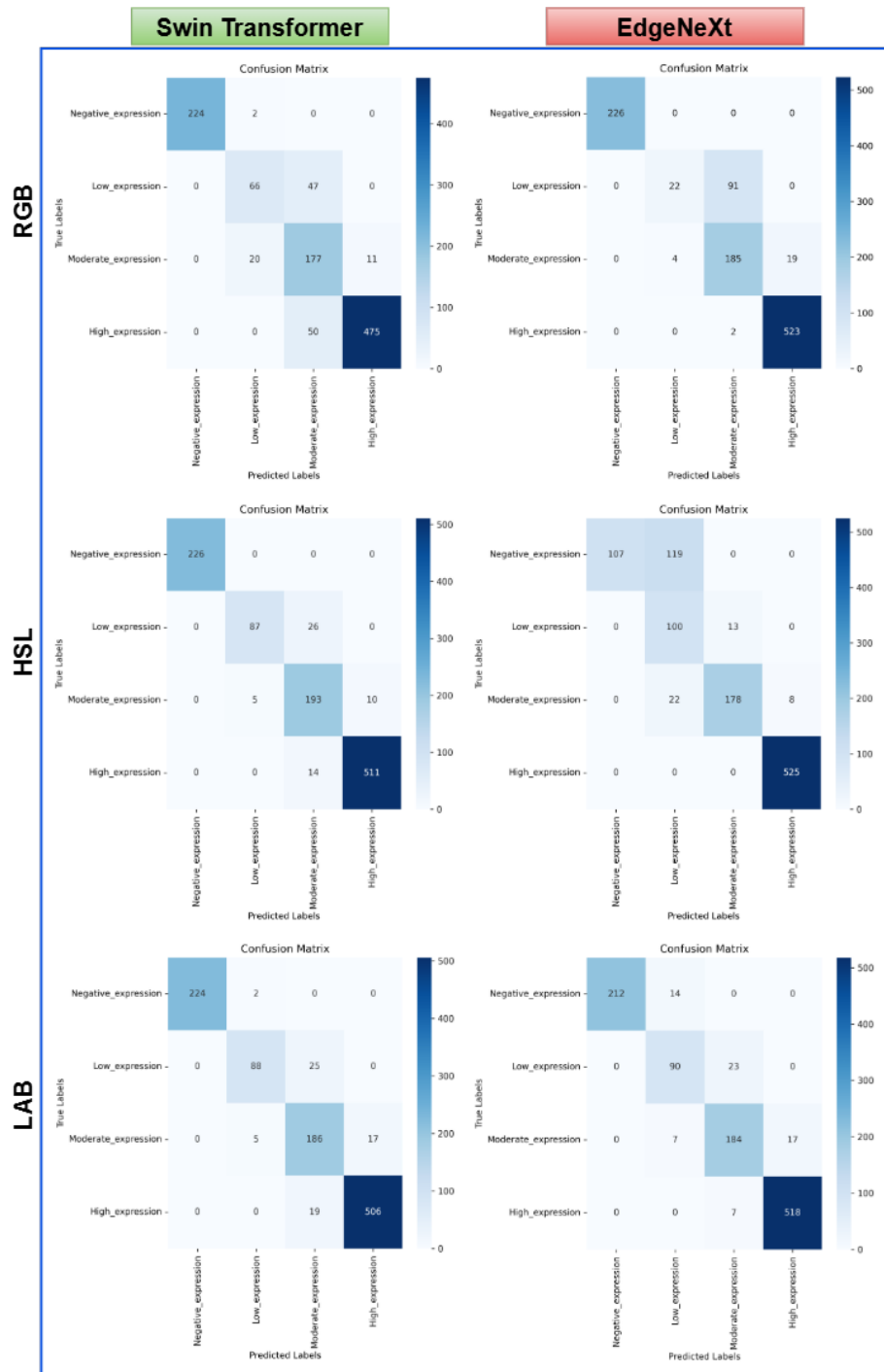


Fig. 4. Confusion matrices for Swin Transformer and EdgeNeXt in RGB, HSL, and LAB color spaces on the OOD Set

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expression, which has a high misclassification rate. However, it performs better across other expression levels and excels in the LAB space. In the RGB space, both Low and Moderate expressions are often misclassified, indicating a decline in classification accuracy. These results suggest that the HSL and LAB color spaces improve model robustness against staining and imaging variations.

4.2 Best Model Selection Using MCDA

Determining the best model and color space combination for IHC ER biomarker classification is challenging due to performance variations across test and OOD datasets. To address this, we employ MCDA using TOPSIS, which systematically ranks models based on these four key metrics across both datasets. Ta-

Table 4. TOPSIS Scores and Model Rankings

Model	TOPSIS Score	Rank
SwnTransformer_HSL	0.9461	1
SwinTransformer_LAB	0.8733	2
EdgeNext_LAB	0.8245	3
SwnTransformer_RGB	0.4044	4
Edgenext_HLS	0.1616	5
Edgenext_RGB	0.1341	6

ble 4 presents the TOPSIS-based ranking, identifying the top-performing model and color space combination. Swin Transformer_HSL achieved the highest TOPSIS score (0.9835), securing the top rank, followed by Swin Transformer_LAB) (0.9613) and EdgeNeXt.LAB) (0.9486), ranking second and third, respectively.

4.3 Ensemble vs. Fusion Performance Evaluation Analysis

This section compares ensemble and fusion approaches using **accuracy, precision, recall, and F1-score** on both test and OOD Sets, along with misclassification analysis to assess localization errors and performance robustness across varying data distributions.

Performance Evaluation and Misclassifications Analysis of Ensemble and Fusion Approach on Test Set and OOD set: Table 5 present the classification performance of Ensemble and Fusion methods on the test and OOD sets. TThe Confidence Weighted Average Ensemble achieves the highest accuracy on both the test set (0.9054) and OOD set (0.9228), demonstrating its effectiveness in leveraging model confidence for improved decision-making. Cross Attention Fusion performs competitively (0.9040 on the test set, 0.9041 on the OOD set), highlighting the benefits of feature integration across multiple

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color spaces. However, ensemble methods outperform fusion-based approaches, emphasizing the advantage of decision-level fusion in enhancing generalization. Figure 5 presents the confusion matrices for ensemble and fusion models on

Table 5. Performance Comparison of Different Fusion and Ensemble Methods

Models	Accuracy	Precision	Recall	F1 Score
Test Set				
Confidence-Weighted Avg. Ensemble	0.9054	0.8946	0.9054	0.8986
Cross Attention Fusion	0.9040	0.8979	0.9040	0.9007
OOD set				
Confidence-Weighted Avg. Ensemble	0.9228	0.9456	0.9228	0.9326
Cross Attention Fusion	0.9041	0.9331	0.9041	0.9158

the test and OOD datasets. The Confidence Weighted Average Ensemble shows high accuracy for Negative, Low, and High Expression classes, with Moderate Expression being the most misclassified on the test set. On the OOD set, it maintains strong performance across all classes. The Cross Attention Fusion model performs well for Negative and High Expression, but struggles with Low and Moderate Expression, especially on the OOD set, where misclassifications are more frequent. Overall, the ensemble model provides better class separation and generalization than the fusion model.

4.4 Grad-CAM Visualization for Model Interpretability.

Figure 6 presents Grad-CAM visualizations for the Two model approaches: Confidence Weighted Average Ensemble, and Cross Attention Fusion. The Confidence Weighted Average Ensemble demonstrates strong localization, effectively focusing on biomarker-expressing regions while minimizing irrelevant activations. In contrast, the Cross Attention Fusion exhibit broader and less concentrated activations, occasionally highlighting non-relevant regions. . In contrast, the fusion model exhibits broader, less concentrated activations, occasionally highlighting non-relevant regions.

5 Conclusion & Future Work

This study evaluated Hybrid Transformer models for IHC ER biomarker Expression classification across three color spaces. MCDA-TOPSIS identified Swin Transformer_HSL, Swin Transformer_Lab, and EdgeNeXt_Lab as top-performing models. The Confidence Weighted Average Ensemble achieved the highest accuracy (90.54% test, 92.27% OOD), demonstrating strong generalization. Grad-CAM visualizations enhanced interpretability by highlighting biomarker-expressing regions. These findings underscore the importance of color space selection and

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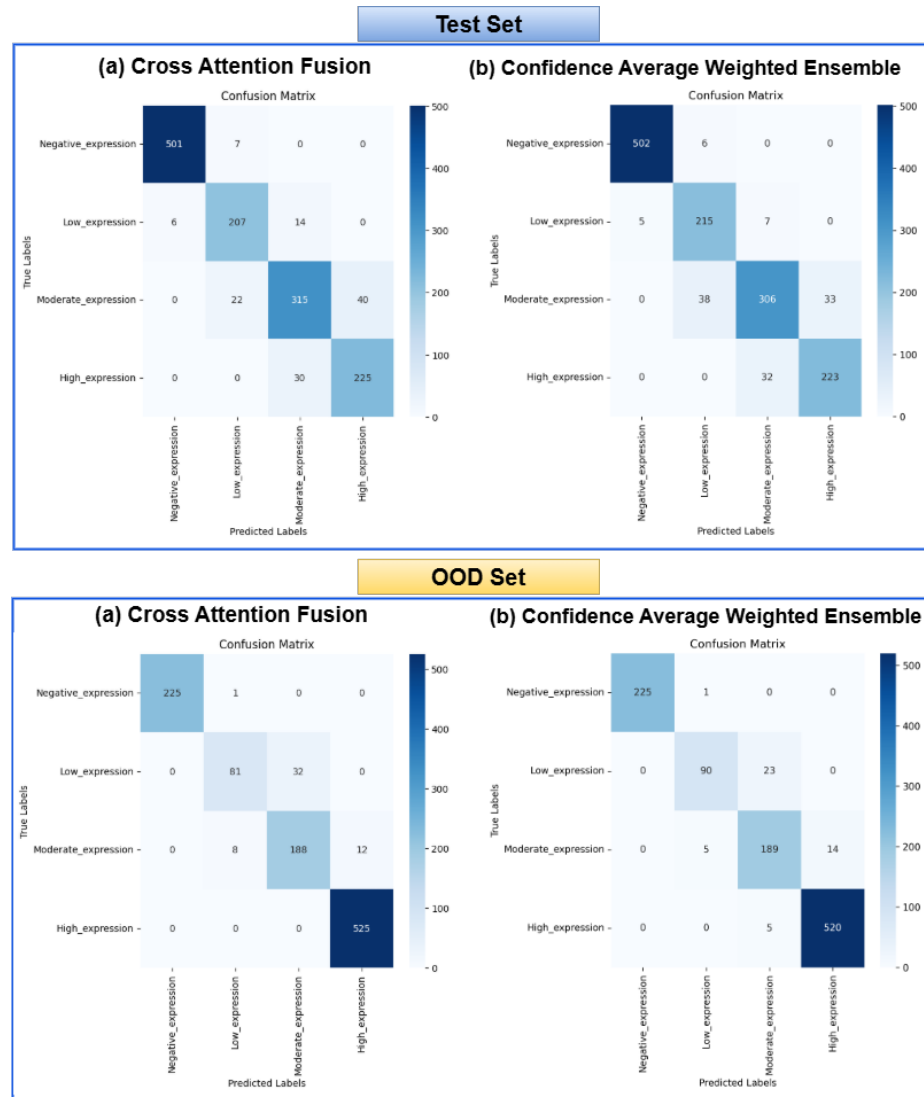


Fig. 5. Confusion matrices for Cross Attention Fusion and Confidence Weighted Average Ensemble on the Test and OOD Set

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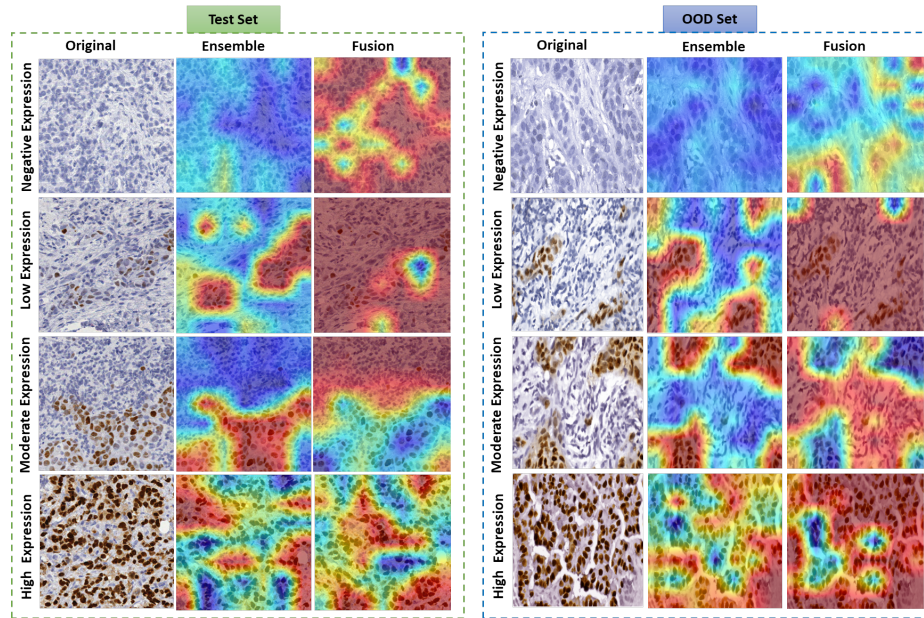


Fig. 6. GradCam Visualization of Ensemble and Fusion on Test and OOD set

model integration. Future work can explore self-supervised learning and domain adaptation to further improve digital pathology applications.

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